

Web-Based PAR Index System

USER MANUAL

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Illustrated with real screenshots from a live end-to-end walkthrough

Abstract

This user manual describes the Web-Based PAR Index System for orthodontic cases. Orthodontists can upload 3D dental scans, score cases against the clinical rubric or use automated scoring, and track outcomes across treatment. Undergraduates can contribute anonymised scans with verified ground-truth PAR scores to grow the machine-learning training dataset, while administrators manage users, oversee submissions, and control the ML model lifecycle.

Index Terms

PAR Index, orthodontics, 3D dental scans, machine learning, user manual.

Document Structure

The manual is organised by system function and user role: system requirements and authentication, undergraduate workflows, orthodontist workflows, administration, troubleshooting, and appendix.

I. INTRODUCTION

The Web-Based PAR Index System automates PAR (Peer Assessment Rating) scoring for orthodontic cases.

Orthodontists upload 3D dental scans, score cases against the clinical rubric (or use automated scoring), and track outcomes across treatment.

Undergraduates contribute anonymised scans to grow the system's ML training dataset. Admins manage users, oversee submissions, and control the ML model lifecycle.

Three roles:

- ORTHODONTIST (full clinical workflow)
- UNDERGRADUATE (training data submission)
- ADMIN (system oversight, pre-seeded account only)

II. SYSTEM REQUIREMENTS

- A modern browser with WebGL support (Chrome, Edge, or Firefox) — the 3D model viewer requires it
- Access to the deployed system's URL, provided by your administrator

III. SIGNING IN AND REGISTERING

A. Login

Enter your email and password, then Sign in.

Fig. 1. Login page.

B. Registration

New Orthodontists and Undergraduates register themselves. Choose your role during sign-up — Admin accounts are pre-seeded only and cannot be self-registered.

Fig. 2. Registration page, showing the two available roles.

IV. UNDERGRADUATE GUIDE

Undergraduates contribute anonymised 3D scan sets with a verified ground-truth PAR score to grow the ML training dataset. Each submission is reviewed and approved or rejected by an assigned orthodontist.

A. Dashboard

Shows your total submissions, how many are pending/approved/rejected, and current ML model status.

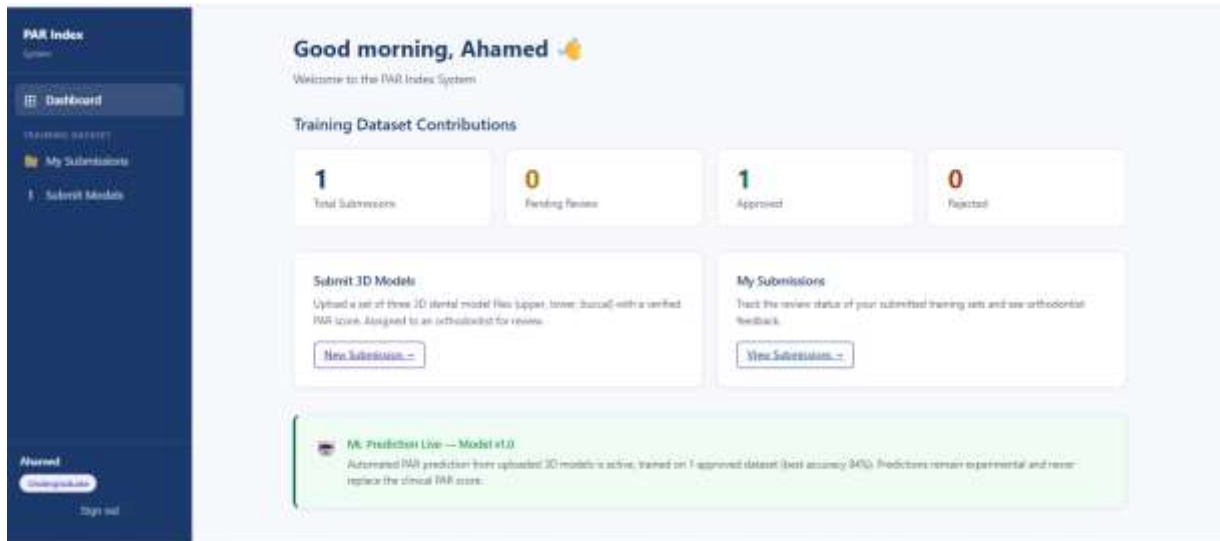


Fig. 3. Undergraduate dashboard.

B. Submitting a training dataset (3 steps)

1. Case Details

give the set an anonymised case label (never a real patient identifier), the ground-truth PAR score verified by your supervising orthodontist, an optional source description, and select the orthodontist who will review it.

2. 3D Model Upload

upload Upper Arch, Lower Arch, and Buccal View files (STL or OBJ, max 50MB each). All three are required.

3. Complete

confirmation that the set was submitted for review.

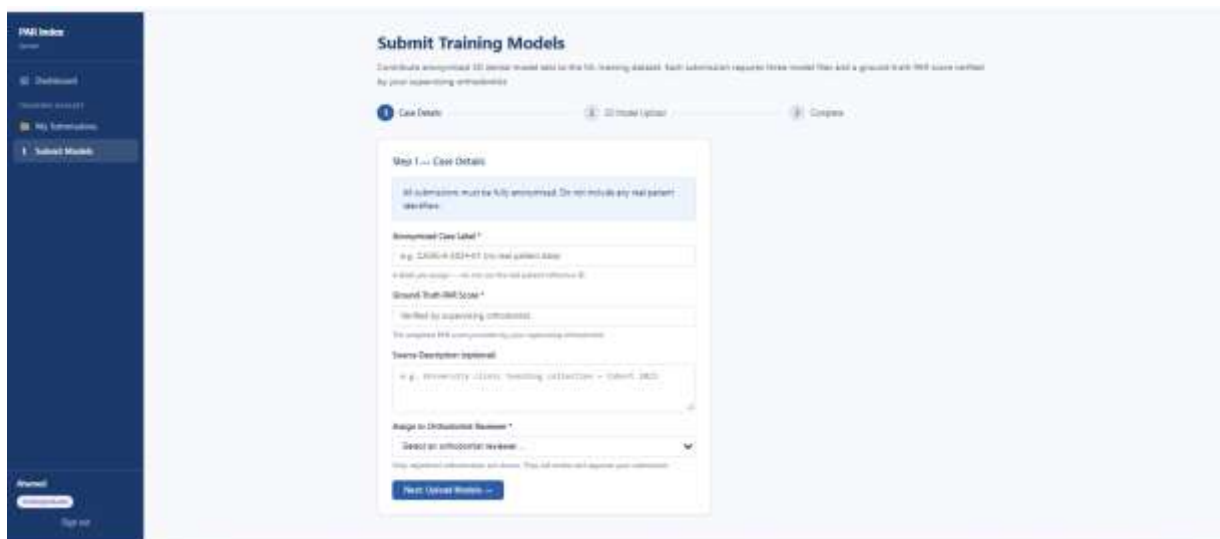


Fig. 4. Step 1: Case Details.

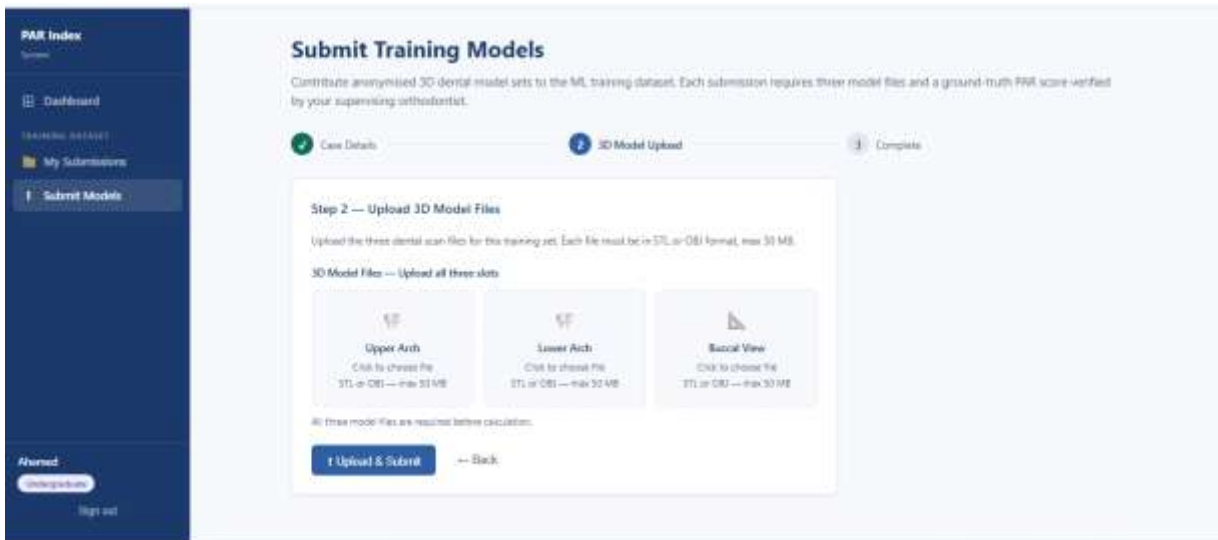


Fig. 5. Step 2: 3D Model Upload.

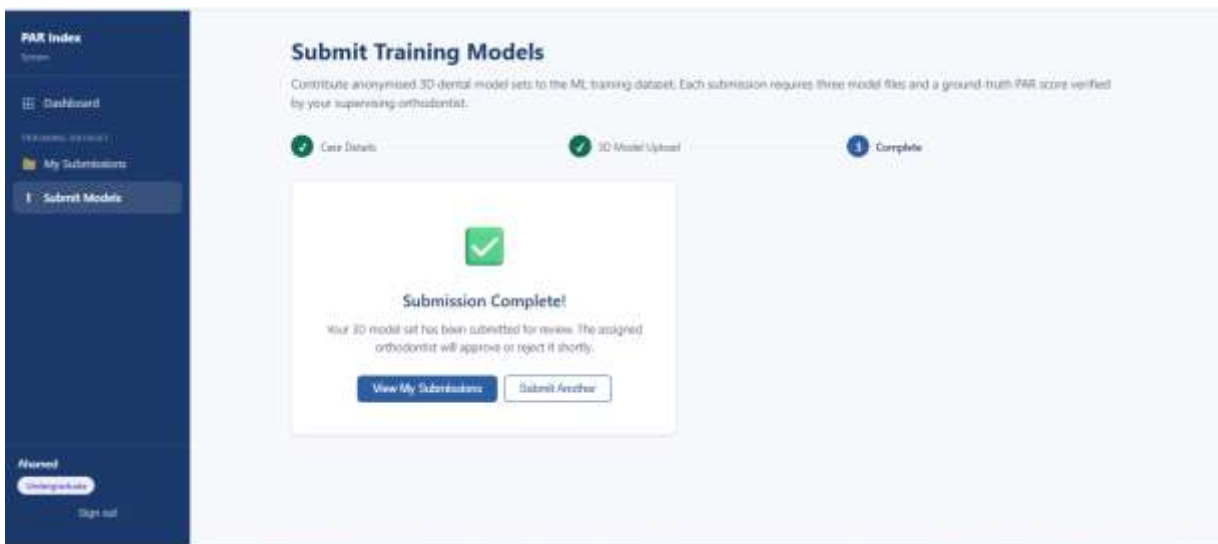


Fig. 6. Step 3: Submission Complete.

C. Tracking your submissions

My Submissions lists every set you've contributed, filterable by status, with the assigned reviewer and any reviewer note. You can delete a submission while it's still Pending.

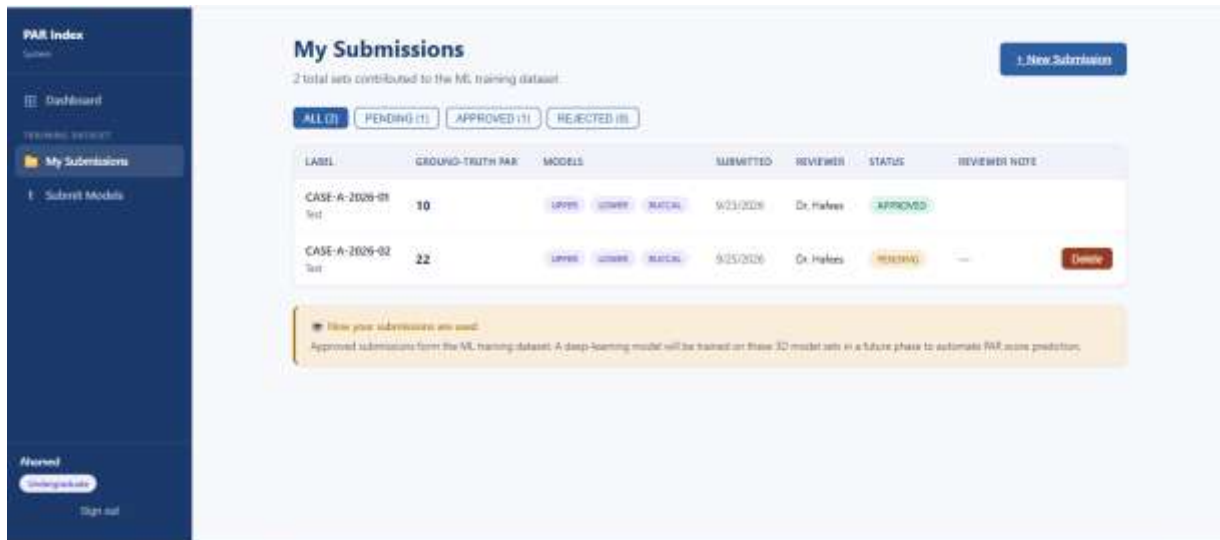


Fig. 7. My Submissions, showing one Approved and one Pending set.

V. ORTHODONTIST GUIDE

A. Dashboard

Shows your patient count, submissions awaiting your review, and current ML model status.

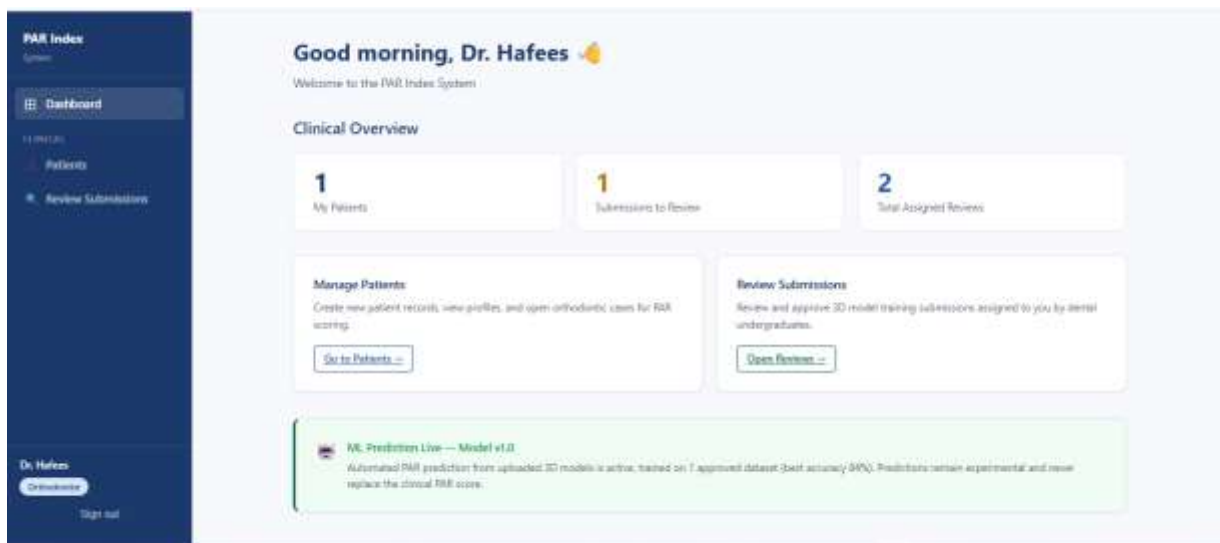


Fig. 8. Orthodontist dashboard.

B. Reviewing undergraduate submissions

Submissions assigned to you appear under Review Submissions.

Opening one shows who submitted it, when, the anonymised label, ground-truth PAR, and the actual uploaded 3D models — viewable arch-by-arch in the same 3D viewer used elsewhere in the system — before you Approve or Reject.

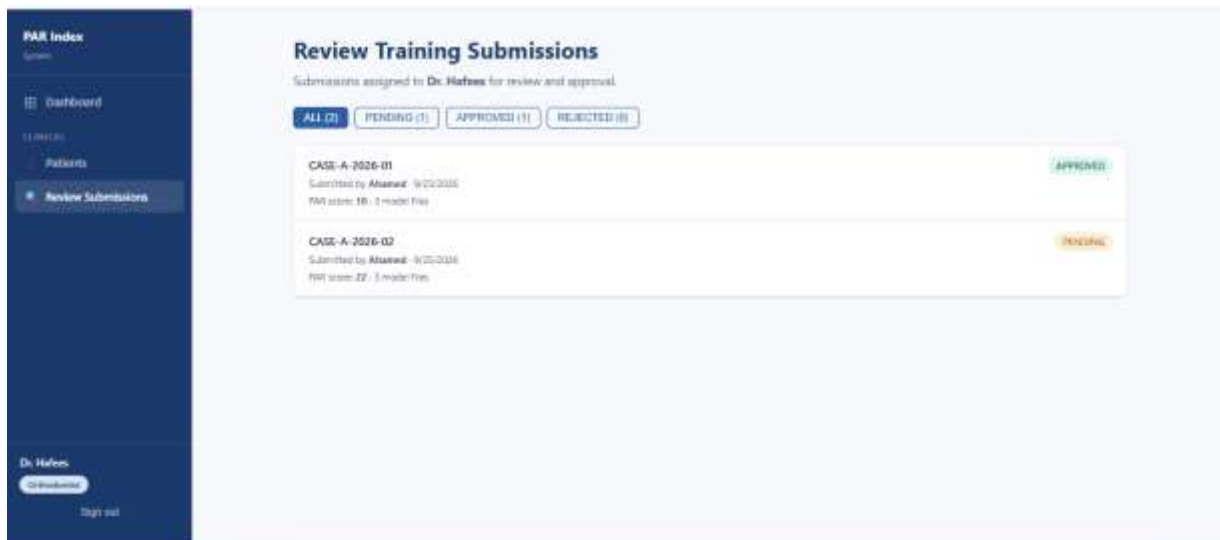


Fig. 9. Review Submissions list.

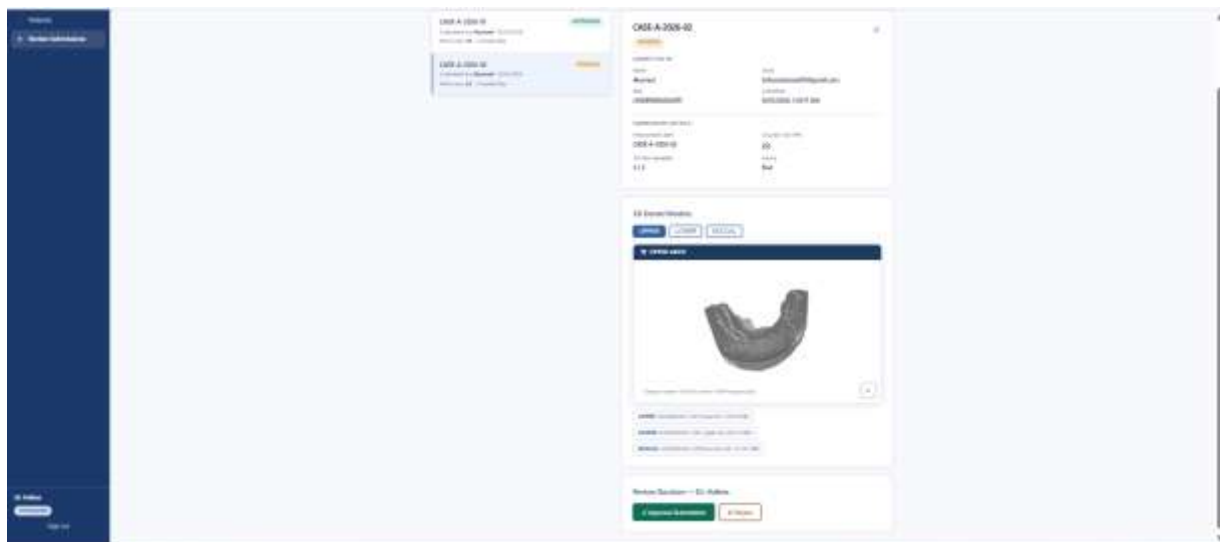


Fig. 10. Reviewing a submission's 3D models before approving.

C. Managing patients

Patients → + New Patient. Reference ID and Full Name are required; Date of Birth and Contact are optional.



Fig. 11. Patients list.

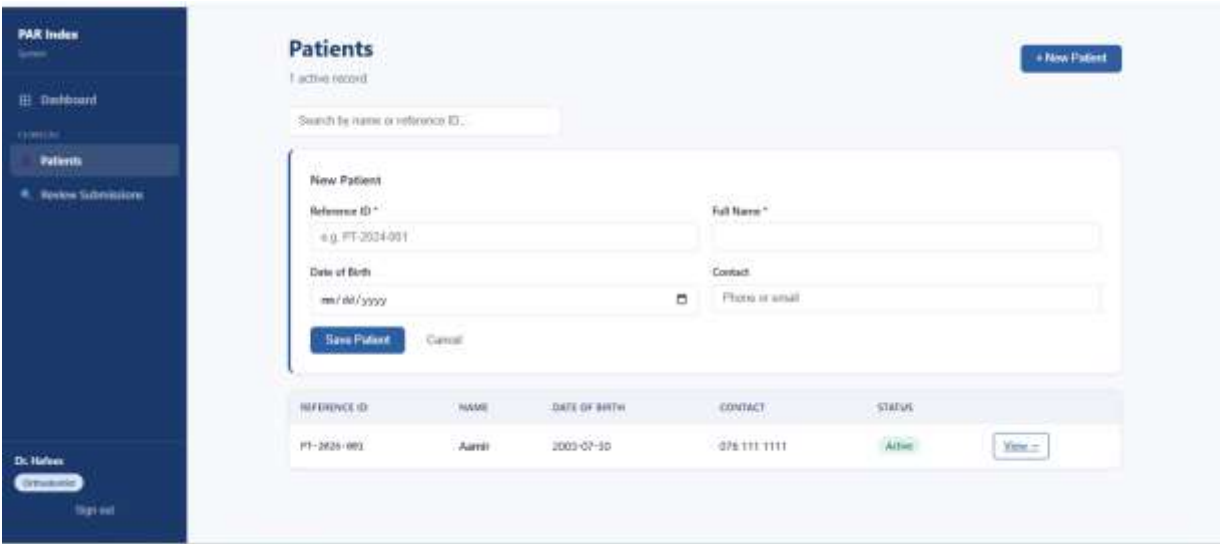


Fig. 12. New Patient form.

D. Creating a case

Open a patient, then + New Case. The first case for a patient is automatically Pre-treatment. Post-treatment only becomes available once a Pre-treatment case for that patient has been finalized — the system then auto-selects Post-treatment and links it to that finalized case for comparison.

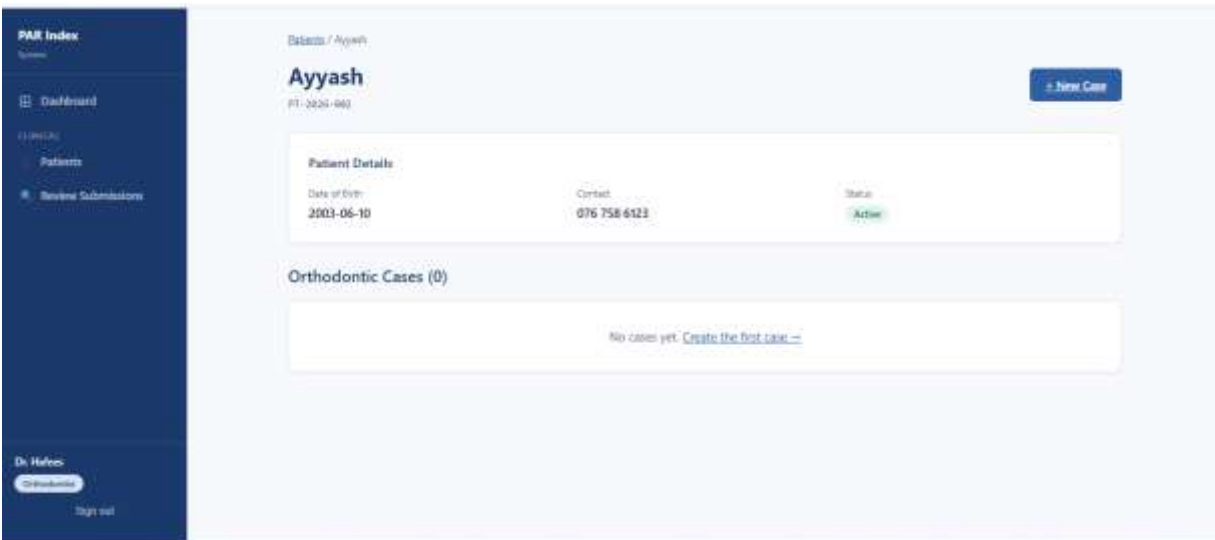


Fig. 13. Patient page, no cases yet.

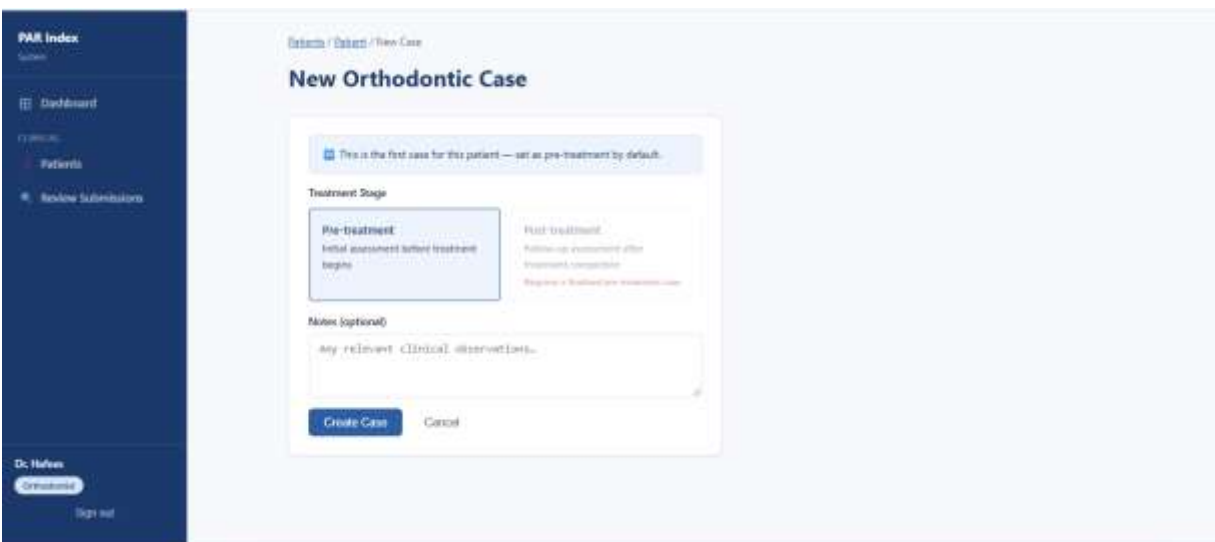


Fig. 14. Creating the first (Pre-treatment) case.

E. Uploading 3D models

Every case needs Upper Arch, Lower Arch, and Buccal View files (STL or OBJ, max 50MB each) before any scoring can happen.

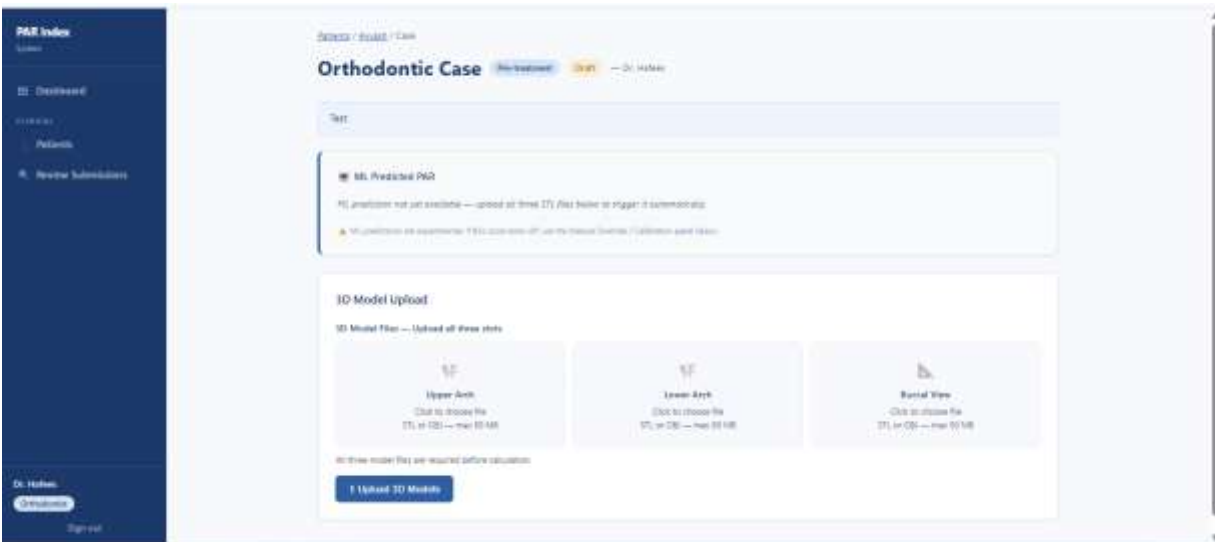


Fig. 15. Case page with the 3D Model Upload panel.

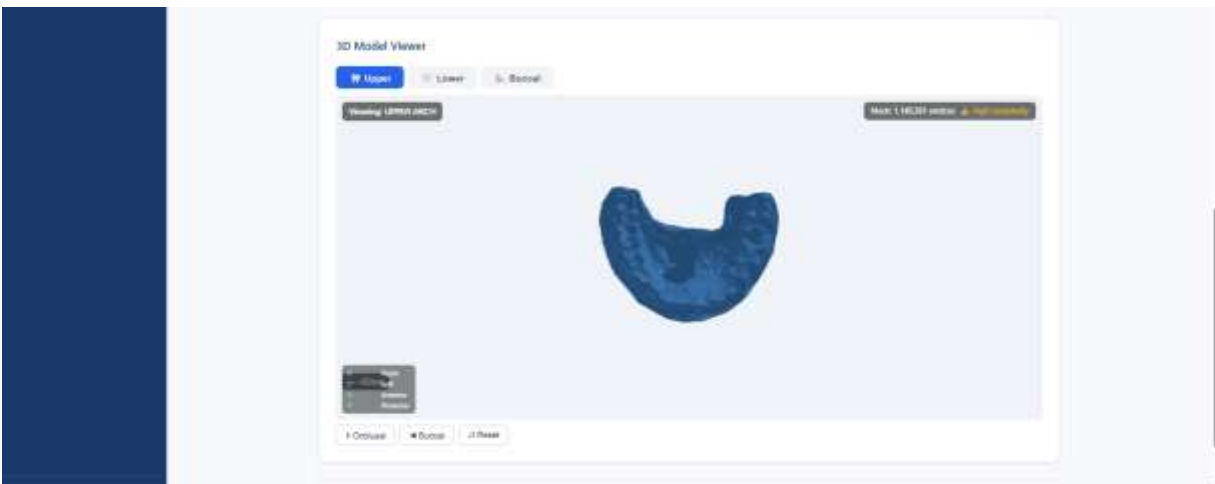


Fig. 16. 3D Model Viewer, showing the uploaded upper arch with mesh info and orientation compass.

F. Scoring a case

The system offers three independent ways to arrive at a PAR score for a case.

You are not required to use all three — most orthodontists only need Manual PAR Scoring.

A. Manual PAR Scoring (the primary method)

Score each component directly against the British PAR Index criteria. The form mirrors the real clinical rubric in detail:

- Upper Anterior Segment and Lower Anterior Segment (×1 each) — five contact points per segment (3-2, 2-1, 1-1, 1-2, 2-3), each scored 0 (0-1mm) upward
- Buccal Occlusion, Left and Right (×1 each) — Antero-Posterior, Transverse (Crossbite), and Vertical (Open Bite), each with its own scale
- Overjet (×6) — first choose Positive Overjet or Reverse/Crossbite, then the millimetre band
- Overbite (×2) — first choose Overbite or Anterior Open Bite, then the coverage band
- Centreline (×4) — a single deviation scale

The weighted total updates live as you score each component; click Calculate PAR Score to save it.

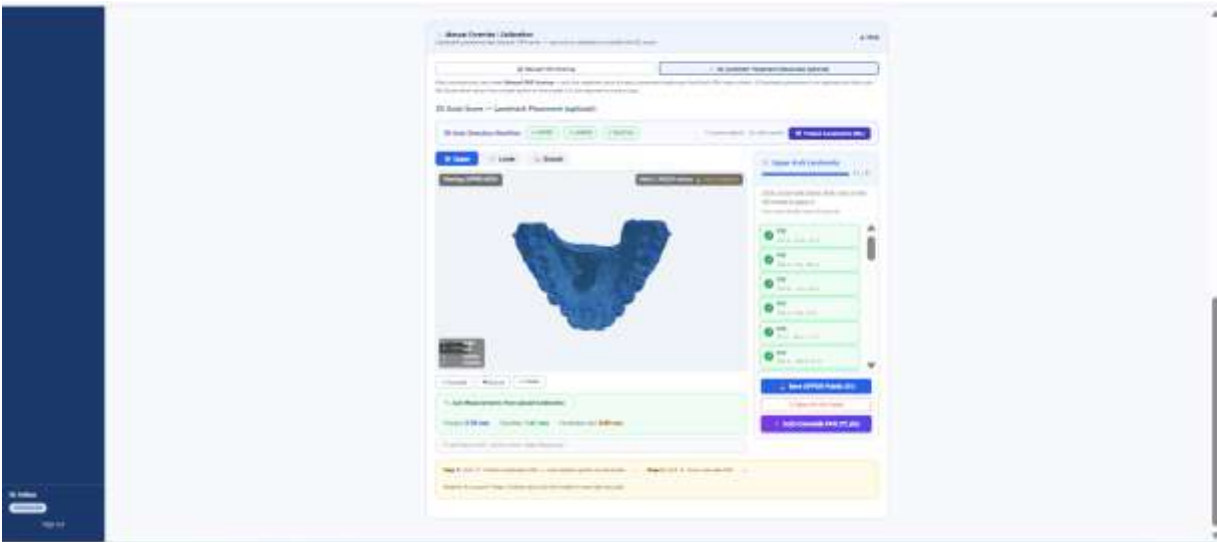


Fig. 19. After Predict Landmarks (ML): all points placed, live measurements shown.

C. ML Predicted PAR (automatic, independent cross-check)

Computed automatically once all three models are uploaded — no manual landmark placement required. Always shown with an explicit experimental-prediction warning; use Use This Score as Final only if you're satisfied it's clinically reasonable.

G. Reviewing the result and finalizing

The case page shows the PAR Score Result (component-by-component breakdown with raw score, weight, and weighted score) alongside the independent ML Predicted PAR, so you can compare them before deciding which to finalize with.



Fig. 20. PAR Score Result (39, from landmarks) next to the ML Predicted PAR (2.5) — shown here to illustrate that the two methods can diverge significantly.

Finalizing locks the case permanently — no further edits are possible without admin intervention. You must type CONFIRM to proceed.

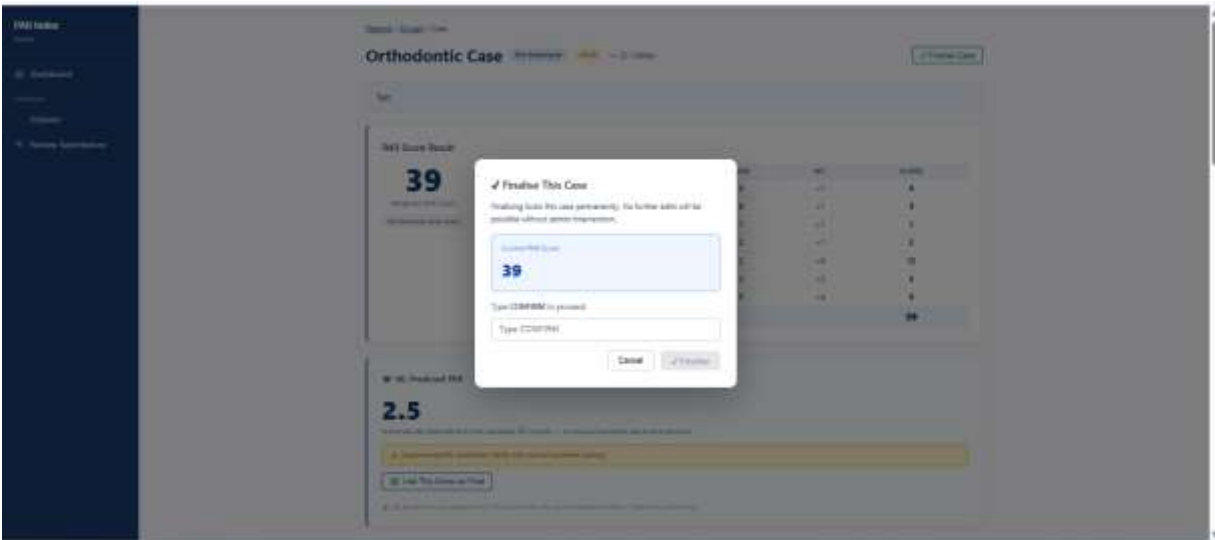


Fig. 21. Finalize confirmation dialog.



Fig. 22. Patient page after finalizing, showing the locked Pre-treatment case.

H. Post-treatment cases and outcome comparison

Once the Pre-treatment case is finalized, creating a new case for that patient auto-selects Post-treatment and shows which Pre-treatment case it will be compared against — double-check this is correct before finalizing.



Fig. 23. New case, auto-set to Post-treatment.

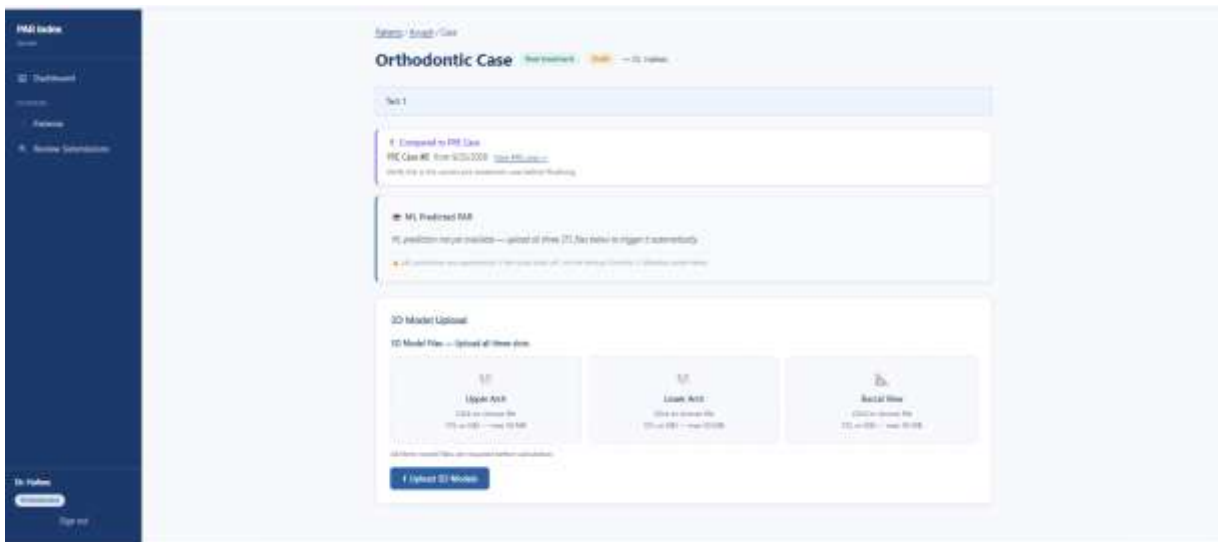


Fig. 24. Post-treatment case, linked to its Pre-treatment case.

The Post-treatment result shows the outcome classification and the point difference against the Pre-treatment score — in this example, a real 14-point increase classified as "No Different or Worse", included here deliberately to show the system reports unfavourable outcomes plainly rather than only favourable ones.



Fig. 25. Post-treatment PAR result with outcome classification.

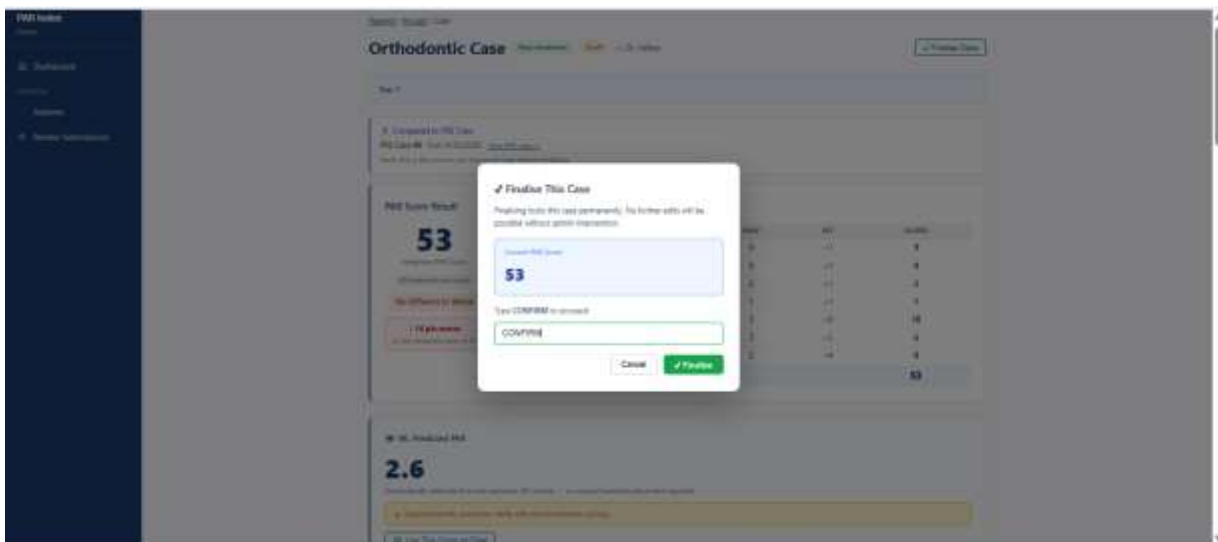


Fig. 26. Finalizing the Post-treatment case.

I. Exporting a PDF report

From a patient's page, Export PAR Report produces a clinical PDF summarising patient information, the pre/post treatment outcome comparison, and the full component-by-component score breakdown for both cases.

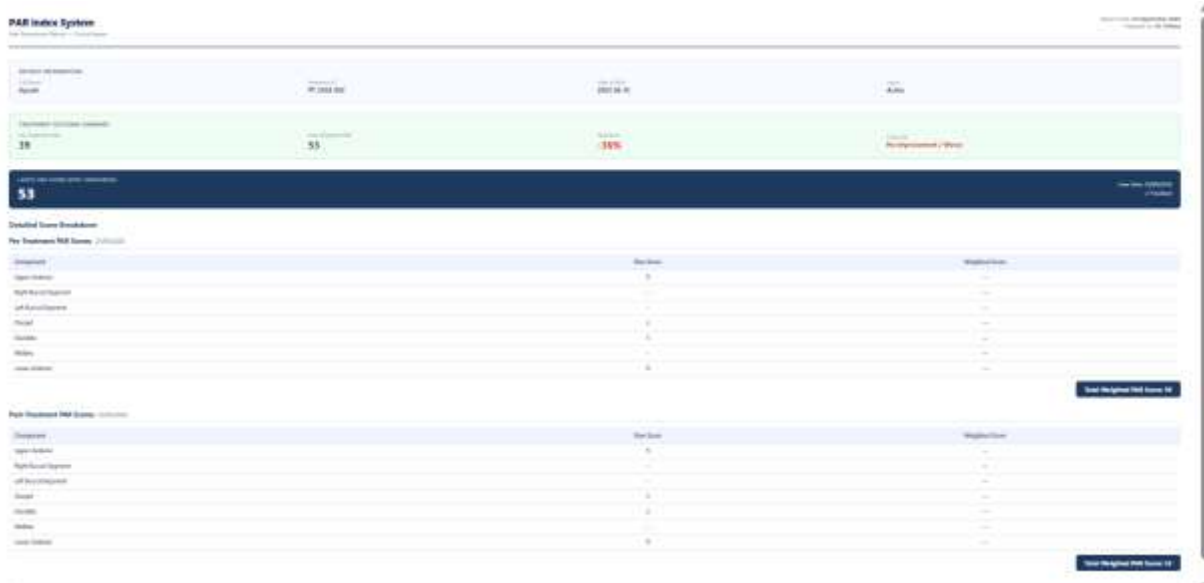


Fig. 27. Exported PAR clinical report.

VI. ADMIN GUIDE

Admin accounts are pre-seeded — sign in with a seeded admin email.

A. Dashboard

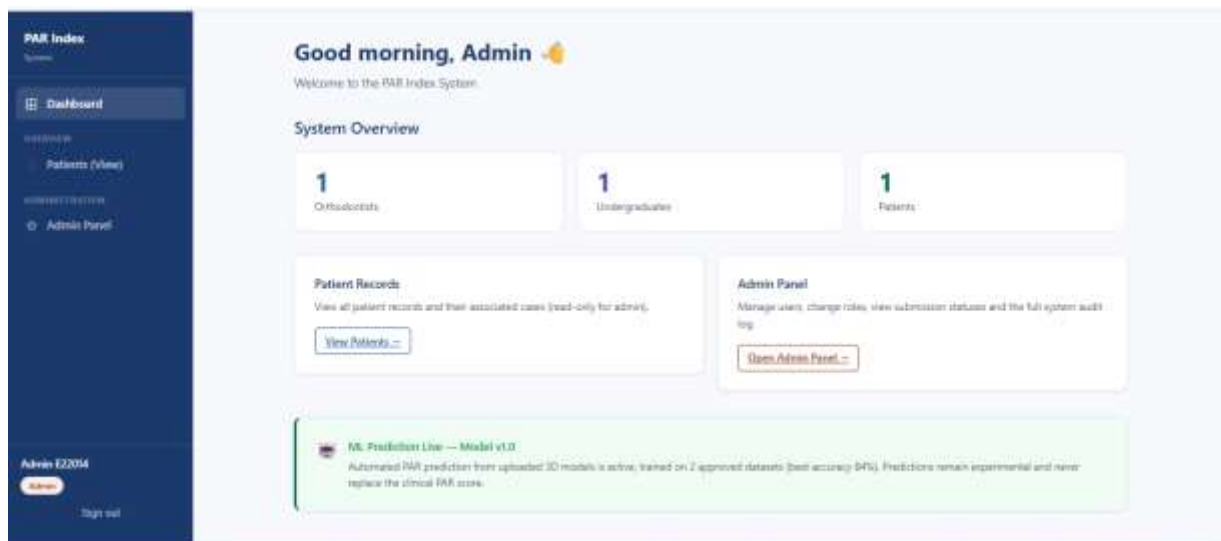


Fig. 28. Admin dashboard.

B. User Management

View all Orthodontists and Undergraduates, change a user's role, and activate/deactivate accounts.

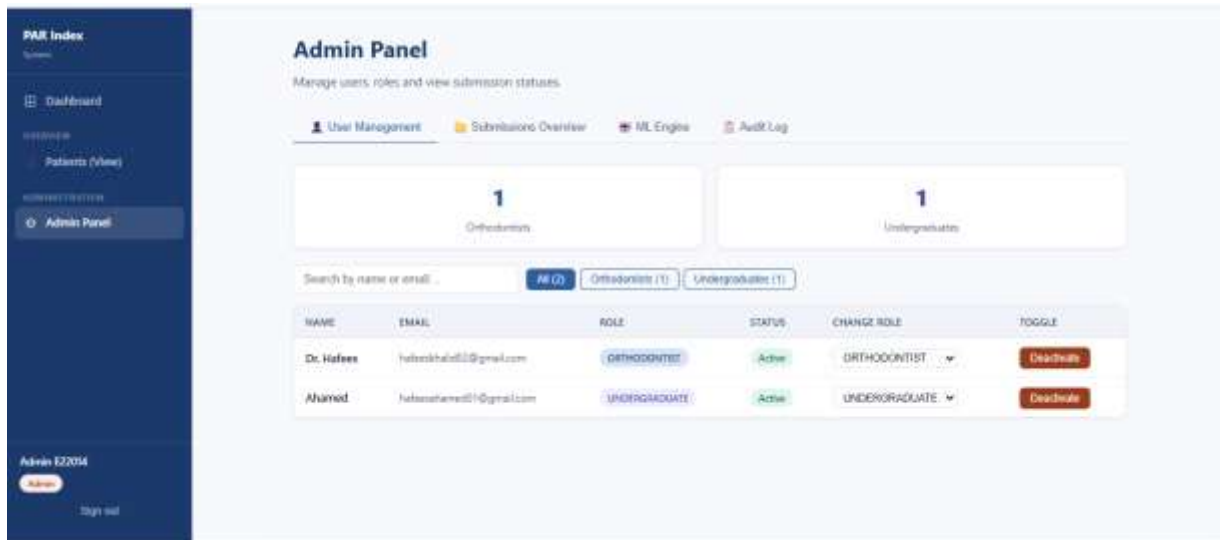


Fig. 29. Admin: User Management.

C. Submissions Overview

A read-only view of all training-dataset submissions system-wide. Approving/rejecting stays with the assigned orthodontist — admins can see status but not act on it here.

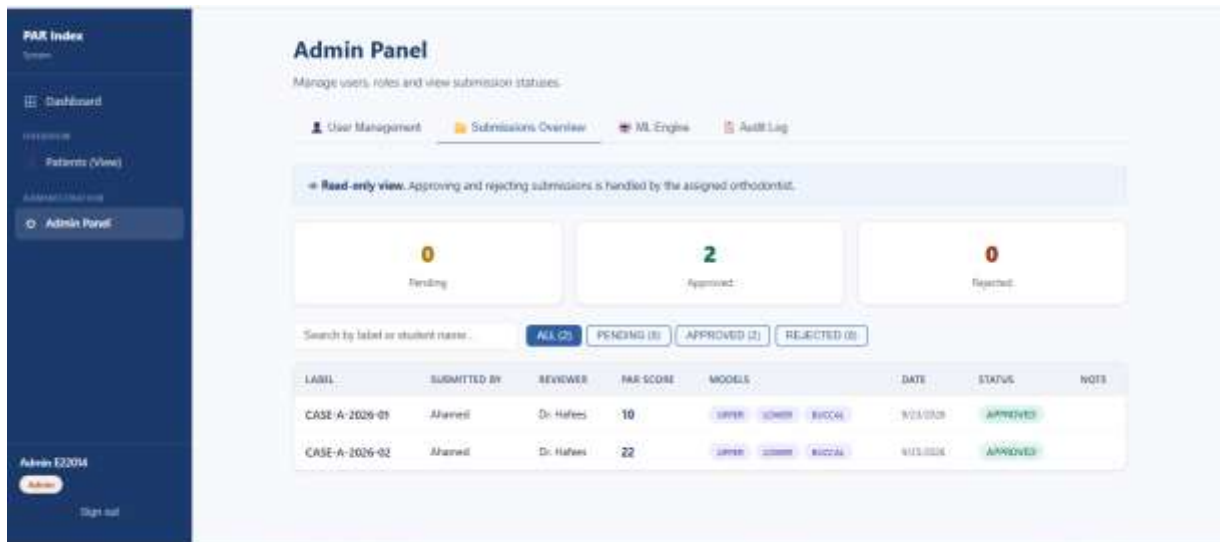


Fig. 30. Admin: Submissions Overview (read-only).

D. ML Engine

Shows current model status, latest version, number of approved datasets, best recorded accuracy, and total training runs. A dataset-confidence warning appears when there isn't enough approved training data yet — the system states 100 approved datasets as the recommended minimum for reliable predictions. Admins can start a new training run (choosing a model version label and epoch count from 10-500) — training runs only on approved datasets, and the previous model file is backed up automatically before each run.



Fig. 31. Admin: ML Engine panel.

E. Audit Log

A complete, searchable, filterable log of every significant system action — logins, case finalization, landmark prediction, PAR calculation (manual, geometric, and ML), with structured detail per entry (e.g. exact scores and model versions involved).

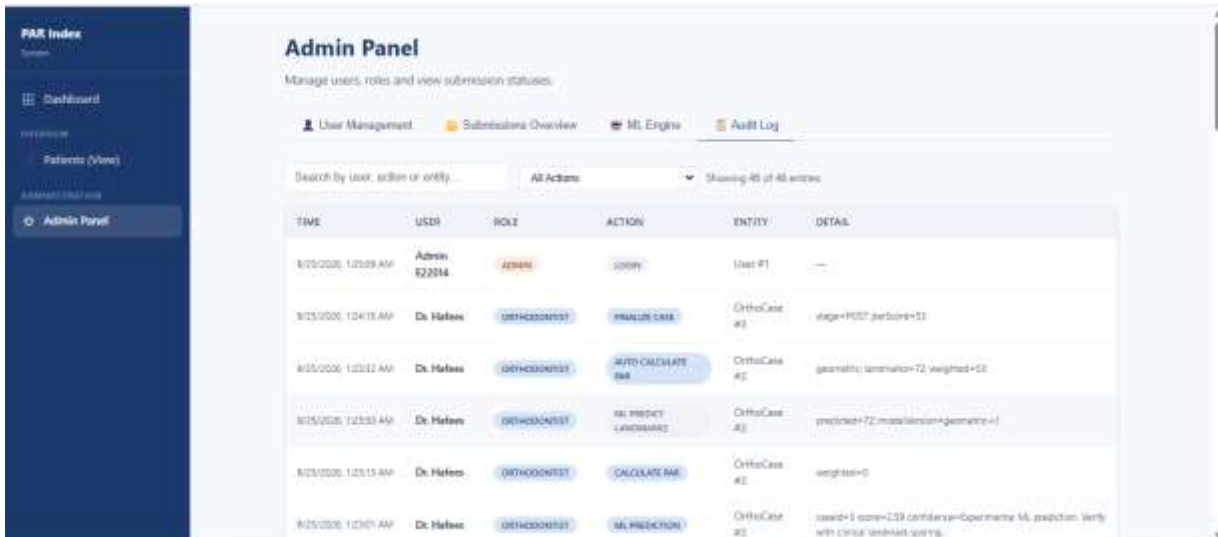


Fig. 32. Admin: Audit Log.

VII. TROUBLESHOOTING

- "All three model files are required before calculation" — upload Upper Arch, Lower Arch, and Buccal View before attempting any scoring method
- Post-treatment option greyed out — the patient's Pre-treatment case must be finalized first
- Training submission won't submit — an orthodontist reviewer must be selected
- Dataset Confidence: Low warning in ML Engine — this is expected until at least 100 approved training datasets exist; it does not block the system from working, it's a caution about prediction reliability
- Need to change a finalized case — this requires admin intervention; finalizing is a deliberate, confirmed action and is not reversible by the orthodontist alone

VIII. APPENDIX

Note: Screenshots in this manual were captured from a real, live walkthrough and reflect the actual implementation at that time. If the UI changes, update the relevant figure.